

Opinion

Improving the Accuracy of Predicted Human Pharmacokinetics: Lessons Learned from the AstraZeneca Drug Pipeline Over Two Decades

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During drug discovery and prior to the first human dose of a novel candidate drug, the pharmacokinetic (PK) behavior of the drug in humans is predicted from preclinical data. This helps to inform the likelihood of achieving therapeutic exposures in early clinical development. Once clinical data are available, the observed human PK are compared with predictions, providing an opportunity to assess and refine prediction methods. Application of best practice in experimental data generation and predictive methodologies, and a focus on robust mechanistic understanding of the candidate drug disposition properties before nomination to clinical development, have led to maximizing the probability of successful PK predictions so that 83% of AstraZeneca drug development projects progress in the clinic with no PK issues; and 71% of key PK parameter predictions [64% of area under the curve (AUC) predictions; 78% of maximum concentration (C_{max}) predictions; and 70% of half-life predictions] are accurate to within twofold. Here, we discuss methods to predict human PK used by AstraZeneca, how these predictions are assessed and what can be learned from evaluating the predictions for 116 candidate drugs.

Development and Alignment of PK Prediction Methods

In response to the recognition that a significant cause of attrition in clinical development of small molecules was due to a lack of appropriate **pharmacokinetic (PK)** (see [Glossary](#)) properties [1], considerable investments were made across the pharmaceutical industry to build the scientific understanding and experimental assays to characterize and predict human PK. This produced a wealth of publications demonstrating **scaling** methodology, from the use of human liver microsomes and hepatocytes for predicting human hepatic clearance (CL) [2–4], to the application of animal *in vivo* data and allometric scaling to predict volume of distribution, CL, and bioavailability [5–7].  modelling also emerged as a mathematical framework describing the major organs and tissues of the mammalian body, connected by the arterial and venous pathways, developed with the intention of: 

 Optimization strategies focused on achieving target potency, selectivity, and an acceptable predicted human PK profile maximize the probability of successful drug discovery. As a compound's key PK properties, such as oral absorption, distribution, susceptibility to being a substrate or inhibitor for drug-metabolizing enzymes (DMEs), and passive and active elimination processes are commonly

Highlights

It is now universally recognized that an acceptable human PK profile increases the probability of a candidate drug becoming a successful therapy.

A variety of tools are available to predict human PK behavior in advance of clinical data, including scaling clearance from *in vitro* metabolic stability data, and physiologically based scaling of volume of distribution.

To improve PK prediction methods, it is crucial to continually assess the performance of predictions with first-time-in-human PK data for new candidate drugs.

To date, AstraZeneca have compared observed PK to predictions for 116 candidate drugs, and since the launch of our five-dimensional framework in 2011, we have driven sustained improvements in the quality of PK predictions.

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associated with inherent **physicochemical properties**, such as molecular size, hydrophobicity, aqueous solubility, and ionization state at physiological pH, understanding these relationships is central to a successful outcome [6].

By the mid-2000s, there was an extensive number of methods available [4,5,8,10,11], but little consensus on how these approaches should define best practice. This led to a cross-industry, precompetitive initiative led by the Pharmaceutical Research and Manufacturers of America (PhRMA)⁵ with the aims of (i) establishing a blinded compound dataset of preclinical and clinical data provided by industry members; and (ii) compiling a list of available scaling methods and testing their accuracy against the dataset to assess the merits of certain data types and scaling methods for predicting human PK. This work was published across five manuscripts [4,5,8,10,11] and provides the most comprehensive assessment to date. The preferred methods that have been used for characterizing, optimizing, and predicting the fundamental human ADME parameters and have laid the foundation of AstraZeneca's PK strategy for small molecules are briefly outlined in the following sections (Figure 1A, Key Figure). There is broad agreement between pharmaceutical companies that the assays described below represent best practice experiments to predict human PK. However, where more than one assay provides information on one of the ADME processes of a candidate drug, they can suggest conflicting pictures of what the properties could be in humans; for example, high or low bioavailability or CL. In this case, it is important not to rely on a single preferred assay or average of results; the AstraZeneca approach is to consider the weight of all the evidence together, along with an understanding of the limitations of each assay, to understand the most likely scenario for the human ADME behavior. AstraZeneca has also developed a method for evaluating the accuracy of human PK predictions against clinical data for candidate drugs, discussed in detail later, which provides learning to refine our interpretation of ADME results in predicting human PK (Figure 1E). The PK prediction assessments also inform the Right Tissue/Exposure component of the AstraZeneca five-dimensional framework, which describes the profiles of risks and opportunities of drug development projects across aspects of Right Target, Right Patients, Right Safety, and Right Commercial Potential alongside the Right Tissue/Exposure [12,13] (Figure 1F). These five dimensions are underpinned by the Right Culture of scientific rigor, and our philosophy of predicting human PK based on mechanistic understanding demonstrates how this culture is embedded in the Right Tissue/Exposure space.

Prediction of Bioavailability and Oral Absorption

Bioavailability (F) (Figure 1B) is the product of the fraction of intestinally absorbed drug (F_a), the fraction escaping intestinal metabolism (F_g), and the fraction escaping liver extraction (F_h). Aiming for a fraction absorbed (F_a) in human of >50% is appropriate, in most instances during the drug screening/selection process, to obtain sufficient bioavailability and ensure that robust systemic exposure to a compound is achieved following oral dosing. Suitable oral drug-like physicochemical properties that deliver both sufficient passive transcellular permeability and aqueous solubility of the crystalline form can lead to confident human absorption estimates of >50%, which can be further strengthened by good *in vivo* absorption, using appropriate formulations, in preclinical species [6,14]. It is necessary to not just determine F in preclinical species, but to deconvolute the contributions of intestinal permeability and efflux and **first-pass** intestinal and hepatic metabolism. Moreover, F , or CL , values in animals are not, *per se*, quality measures to predict F or CL for humans; however, low bioavailability in preclinical species may create obstacles in obtaining appropriate exposures in pharmacodynamic (PD), efficacy, and safety experiments for a candidate drug.

To inform on the potential for oral absorption, the **apparent transcellular permeability** (P_{app}) measure is obtained from an assessment of drug flux across a monolayer of cells with

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characteristics of the intestinal barrier, typically Madin–Darby canine kidney (MDCK) or colorectal adenocarcinoma-2 (Caco-2) cells [15,16]. The absorption of drugs with good permeability and solubility is unlikely to be limited by efflux via P-glycoprotein (an efflux transporter present in the gut and liver) or other intestinal drug transporters unless the dose is low [5,17]. This should not necessarily be assumed if the maximum flux (J_m) for the transporter is high and/or the solubility or permeability is marginal. For compounds that have suboptimal properties, dynamic simulations using either proprietary (GI Sim) or commercial [Simcyp (Certara)ⁱⁱ, Gastroplus (Simulations Plus)ⁱⁱⁱ] software are useful for hypothesis generation and decision making [9,18–20].

Approaches to predict F_g are being developed [21]. The major human intestinal DMEs are **cytochrome P450 (CYP)3A**, CYP2C9 and uridine 5'-diphospho-glucuronosyltransferase (UGT) [22–27]. Intestinal extraction is typically associated with metabolically unstable drugs, yet the enterocyte location of CYP3A4 and P-glycoprotein, which share substrate specificity, facilitates cycling of drugs, prolonging exposure and increasing potential for intestinal metabolism [15,28,29]. Therefore, optimization of hepatic CL , efflux, and permeability properties will reduce factors contributing to intestinal metabolism and mitigate against intestinal metabolism being a significant issue [29].

Prediction of Volume of Distribution

Volume of distribution (V_d) (Figure 1B) is a measure of the relative affinity of a drug for tissues and plasma. Generally, physicochemical properties rather than specific pharmacophores regulate V_d values: acidic drug-like molecules have low (<1 l/kg); neutrals and zwitterions moderate (1–3 l/kg); and basic drugs high (>3 l/kg) V_d values [6,14,30,31]. The dominant influence on the distribution of acidic drugs is the extensive binding to plasma albumin, and they are thus associated with similar distribution volumes to plasma (0.1–0.3 l/kg) unless active transport processes contribute to increase tissue concentrations; for example, via organic anion transporter polypeptide (OATP) [32]. Bases, being positively charged at physiological pH, interact with anionic phospholipid head groups and subcellular acidic organelles; for example, lysosomes, leading to higher tissue affinity and distribution [6,33]. Given the dependency on physicochemical properties, V_d is not a parameter that is readily adaptable in a given chemical series; rather, boundary estimates of V_d , and specifying a desired human elimination half-life ($t_{1/2}$) can help identify a target CL value to optimize against.

V_d is reasonably predictable from physicochemical properties, *in silico* [34,35] and *in vitro* methods [36,37]. Using preclinical *in vivo* PK data and correcting for species plasma protein binding (PPB) differences, V_d is the most predictable of PK parameters; if unbound volume of distribution (V_u) is consistent across species (it is typically conserved within twofold), human V_d can be predicted from animal data by estimating V_u [6,14,31,38]. As PK from multiple species is determined in drug discovery, this is the preferred predictive methodology in AstraZeneca for candidate drugs.

Prediction of CL

The major drug CL (Figure 1B) routes in humans and preclinical species are hepatic metabolism, and renal and **biliary elimination**, and methods are required to predict these parallel elimination processes in humans, using a combination of animal data and *in vitro* human tools. Identification of the elimination route and rate in preclinical species and optimization of human CL (most commonly hepatic) are among the foremost challenges of many drug discovery projects. The liver is the major drug-metabolizing organ, and ***in vitro*–*in vivo* extrapolation (IVIVE)** can be used to estimate hepatic metabolic CL . *In vitro* turnover rates determined using hepatocytes, or subcellular liver fractions, such as microsomes, are scaled to predict *in vivo* liver CL using a

Glossary

Absorption: process by which a drug compound transfers from an extravascular site of dosing (e.g., gut, lung, muscle, skin) into the systemic circulation.

Active transport: transfer of a drug compound across a membrane against the concentration gradient.

Allometry, allometric scaling: estimating the unknown PK parameters (e.g., clearance, volume) in humans from measured values in one or more preclinical species by assuming a relationship between the PK parameters and body weight.

Apparent transcellular permeability (P_{app}): permeability across a cell membrane measured in an *in vitro* system.

Area under the curve (AUC): time integral of a concentration versus time profile, which provides a measure of total exposure, incorporating both amount and duration of the drug concentration.

Ascending dose trial: early clinical trial of a candidate drug, in which increasing doses of the candidate drug are investigated to understand its safety and PK.

Biliary elimination: removal of a drug compound or metabolite from an organism via the bile, leading to its excretion in feces.

Bioavailability: fraction of drug compound dosed by an extravascular route that reaches the systemic circulation.

Clearance (CL): parameter that describes the rate of transformation or removal of a drug compound from circulation or an organ, expressed as the ratio of the loss rate of amount of drug to the instantaneous drug concentration.

Conjugative route of metabolism: chemical modification of a compound by covalent attachment of a chemical group; for example, glucuronic acid.

Cytochrome P450 (CYP): group of DMEs that are involved in metabolism of most drugs.

Disposition: combination of absorption, distribution, metabolism, and excretion processes that describe the fate of a drug compound as it passes into, around, and out of an organism.

Dissolution rate: rate at which solid drug material goes into solution (in the gut or lungs).

Distribution: processes by which a drug compound travels to and reversibly binds to body tissues.

mathematical model [2,3,39–43]. The model describes hepatic metabolic CL (CL_h), via liver blood flow (Q_h), **hepatic intrinsic clearance** (CL_{int}) and **fraction unbound in blood** (f_u).

Both hepatic microsomes and hepatocytes are used in AstraZeneca; hepatocytes contain the full complement of DMEs, while microsomes contain membrane bound DMEs such as CYPs. The method involves determination of hepatocyte (or microsomal) intrinsic clearance rate (CL_{int}) and fraction unbound in the incubation ($f_{u,inc}$), to calculate unbound *in vitro* CL_{int} , which can be scaled up to a whole liver unbound CL_{int} . Evaluation of the predictive accuracy can be made against the relationship of unbound *in vitro* versus *in vivo* CL_{int} , in animals for the chemical series of interest and in human for drugs with clinical PK. This transformation of unbound *in vitro* to *in vivo* CL_{int} yields a systematic underprediction of *in vivo* CL_{int} , which can be corrected empirically via use of a multiplicative factor or regression line, and, importantly, appears across different laboratories, species and hepatocytes/microsomes. In our experience, the empirical correction can be used for substrates of CYPs, flavin-containing monooxygenase (FMO), and some conjugative routes of metabolism, and is periodically reassessed via a test set of compounds [2,39,44,45].

Renal elimination may involve both **passive transport** and active transport components. The renal filtration rate, as a CL , is a product of a drug's fraction unbound in blood (f_u) and glomerular filtration rate (GFR, ~ 1.7 ml/min/kg in humans). As the majority of drugs are lipophilic ($\log D > 0$), and therefore highly plasma bound and passively reabsorbed from the urine, passive renal CL is usually a minor elimination route. Active renal secretion consists of uptake across the basolateral membrane of the proximal tubule of the kidney, then transport across the apical membrane via transporters polarized to the apical or basolateral membrane. In our laboratory, animal renal CL , corrected for PPB and kidney blood flow species differences, provided a reasonable prediction for a diverse set of actively transported drugs, and forms the basis of our approach to predicting renal CL in human [46]. The development of physiologically relevant *in vitro* models for the prediction of human active renal CL is ongoing [47,48].

Biliary excretion of drugs can be a two-step process involving the hepatic uptake and bile canalicular efflux transporters. In our laboratory, human biliary CL is tentatively estimated from the dog or rat biliary *in vivo* CL for non-OATP substrates, since a comprehensive assessment of available clinical and preclinical data indicates that drugs lacking active hepatic uptake have minimal differences in rat, dog, and human biliary CL , whereas large differences can be expected for OATP substrates [49]. Consequently, human-derived *in vitro* models are in active development for the prediction of biliary CL and used for mechanistic investigatory purposes.

For candidate drugs, a detailed quantitative mechanistic understanding of routes and rates of elimination pathways in animals and human *in vitro* systems is critical, in our opinion, to establish confidence in an accurate human CL prediction, and this approach is a foundation of our strategy. This systematic approach requires intellectual and financial investment for careful tailored experimentation, which complements a routine screening cascade, common to many pharmaceutical companies, that may be adequate for, but somewhat limited to, identification of an encouraging chemical space or shortlist of promising compounds [50–52].

Allometry is still widely used within the industry to predict human CL . Publications from peer pharmaceutical laboratories [53–56] confirm the authors' view that allometric scaling approaches offer poor predictivity of human CL ; in particular for metabolic CL , where there may be clear species differences in metabolic rates and routes. However, in the absence of evidence indicating clear species differences in **disposition**, an allometric prediction in agreement with our mechanistic and validated approaches can increase confidence, thus adding value.

First-pass (metabolism): first pass over liver is the fraction of the drug that escapes the liver without metabolic transformation when the drug is transported from the intestinal tract via the portal vein and through the liver into the systemic circulation.

Fraction unbound in blood (f_u): fraction of drug concentration that is free from reversible binding to proteins and cells in blood.

Fraction unbound in the incubation ($f_{u,inc}$): free fraction of compound in an *in vitro* metabolic stability assay (with hepatocytes or liver microsomes), used as part of scaling to predict *in vivo* clearance by relating free concentrations between the *in vitro* and *in vivo* systems.

Glomerular filtration rate (GFR): filtration of drug compounds out of circulation and into the glomerular capillaries in the kidney, leading to renal excretion.

Half-life ($t_{1/2}$): time taken for the drug concentration to decrease to half its initial value. It is related to the reciprocal of the elimination rate, or the ratio of volume of distribution to clearance.

Hepatic intrinsic clearance (CL_{int}): maximum rate at which the liver can metabolize a compound, if blood flow delivering the compound to the liver is not rate limiting.

Hepatocytes: liver cells that are capable of metabolizing drug compounds, and can be cultured *in vitro* for metabolic stability assays.

***In vitro*–*in vivo* extrapolation:** relationship between metabolic clearance rates *in vitro* (from hepatocytes or liver microsomes) and *in vivo*, which can be used to scale an *in vitro* drug turnover rate to predict an *in vivo* hepatic clearance value.

Log D: logarithm of the distribution coefficient (D) describing the partitioning of a compound between lipid and water; higher log D means a more lipophilic and hydrophobic compound.

Lysosomal trapping: sequestration of a basic compound within lysosomes, due to the compound in an unionized state outside the lysosome being able to enter across the membrane, then being unable to pass back through the membrane to exit after being protonated in the acidic environment inside the lysosome.

Maximum concentration (C_{max}): the maximum concentration observed in a concentration–time profile.

Microsomes: subcellular fraction of hepatocytes, derived by centrifugation,

Prediction of Pulmonary Absorption and PK

Direct delivery of drugs to the lung promotes high local tissue concentrations, which drive therapeutic effect, whilst facilitating lower systemic exposure [57]. Rapid absorption from the airways into systemic circulation is expected for small molecule drugs and other therapeutic agents, such as peptides and proteins [58,59]. Consequently, inhaled drugs designed to treat pulmonary disease require optimization of molecular structure and/or formulation to be retained in the lung for a sufficient period of time to allow for a sustained local effect. The two main strategies for optimizing lung retention after inhaled administration of drug therapy involve maintaining a reservoir of drug in the airways, through low solubility and slow **dissolution rate**, or in the lung tissue through the PK distribution properties of the drug [58]. The former case is exemplified by the inhaled corticosteroids and the latter by bronchodilators, where longer $t_{1/2}$ in lung has been achieved by increasing the overall basicity of the molecules, giving higher tissue affinity through mechanisms such as **lysosomal trapping** [60].

Once inhaled particles are delivered into the trachea and deeper airways, subsequent PK profiles can be influenced by a number of factors, including the form of the material (e.g., salt form and particle size profile), the inherent physicochemical properties of the compound, and the influence of disease on the integrity of the lung. The heterogeneous nature of the lung and the range of processes affecting pulmonary drug disposition, as discussed above, make prediction of PK of inhaled drugs challenging. Prior to pulmonary absorption and/or target interaction, deposited particles have to dissolve in the epithelial lining fluid, which covers the epithelium. Undissolved particles are transported towards the pharynx by mucociliary clearance (MCC) and eventually swallowed. The **MCC transport rate** varies across the lung with a maximum velocity in the trachea, but typical half-lives for the overall CL are in the range of 0.5–2 h [61]. Absorption rates from conducting airways to central (bronchial) blood and from small airways/alveoli to peripheral (pulmonary) blood vary due to epithelial depth differences (60 μm in the central airways to less than 3 μm in the small airways/alveoli) and differences in the flow of the two blood supply routes [62]. All of these aspects have a role to play in defining the lung PK profile and, therefore, can influence the overall therapeutic effectiveness of the inhaled medicine. Inhalation-specialized PBPK models have been developed at AstraZeneca to address these complexities (Box 1). These models have been shown to effectively predict not only preclinical lung and systemic PK but also clinical PK and target engagement biomarkers, demonstrating the ability to predict unbound drug concentration at different target locations in the lung [63–65].

Assessment of Prediction Accuracy

Once clinical PK data are available for a candidate drug, the observed human PKs can be compared with predictions, providing an opportunity to assess and refine prediction methods (Figure 1E). The primary PK parameters F , V_d , and CL (Figure 1B) are critical for PK prediction because they are mechanistic parameters that relate to the fundamental ADME processes. The V_d and CL in human can be measured from a PK profile following intravenous (IV) dosing [4,10] and, if an extravascular route (e.g., oral, inhalation) is intended for the candidate drug, F can subsequently be determined by comparing the PK following extravascular dosing with that from IV. For many extravascular administration drugs, however, IV dosing is not explored in the clinic, and we can only determine the apparent extravascular V_d and CL , V_d/F , and CL/F respectively [4,10]. Integrative modelling methods such as PBPK modelling (Figure 1C) are used to predict the PK profile over time in humans, based on the predicted values of F , V_d , and CL . Secondary PK metrics, such as **area under the curve (AUC)**, **maximum concentration (C_{max})**, and $t_{1/2}$, can be derived from the predicted profile (Figure 1D), and provide three metrics that can be compared with observed clinical PK data regardless of the route of administration. The relationships between F , V_d , and CL and AUC, C_{max} , and $t_{1/2}$ can be complex, especially

containing a high proportion of membrane-bound drug metabolizing enzymes such as CYP.

Mucociliary clearance (MCC)

transport rate: physiological rate of movement of mucus by the cilia in the lungs, which affects the clearance rate of compound within the mucus.

Passive (transcellular) transport: transfer of a drug compound across a membrane by diffusion or filtration.

Pharmacodynamics (PD): The time dependence of the pharmacological effects of a drug.

Pharmacokinetics (PK): The time dependence of drug concentration due to the combined effects of absorption, distribution, metabolism, and excretion.

Pharmacophore: chemical substructure that confers target pharmacological activity on a drug molecule.

Physicochemical properties: chemical-specific properties such as lipophilicity, solubility, polarity, and molecular weight, which can affect ADME characteristics of the chemical.

Physiologically based pharmacokinetic modelling (PBPK): a mathematical framework for describing the disposition of a drug, in which the tissues and vascular compartments where drug can be present are represented with physiologically realistic volumes, and connected with physiologically-realistic blood flows.

Plasma protein binding (PPB): reversible binding of a drug compound to soluble proteins in plasma.

Scaling: extrapolation methods used to predict PK parameters (e.g., clearance, volume) in humans from *in vitro* or *in vivo* preclinical systems.

Turnover rate: rate of loss of a drug compound in an *in vitro* metabolic stability assay (with hepatocytes or liver microsomes).

Unbound volume of distribution (V_d): the volume apparent to the free concentration, which is typically similar across species for a given compound.

Volume of distribution (V_d): a PK parameter that describes the extent of a drug compound's distribution to tissues.

for PK behavior with multiple phases or nonlinearity, but in general, AUC is inversely dependent on CL/F ; $C_{\max}$ increases if F increases or V_d or CL decreases; and $t_{1/2}$ increases if V_d increases or CL decreases.

The accuracy of human PK predictions is usually assessed by calculating the ratio of predicted to observed secondary PK parameters [4,5,8,10,11,66–68]. In the largest analysis of this type, we have reviewed 116 candidate drugs in clinical development between 2000 and 2018 (Table 1), allowing for subset analysis, for example, by date or chemical class. Date partitioning at 2011 was performed to evaluate recent progress in prediction performance versus historical performance, and also aligned with implementation of the AstraZeneca five-dimensional framework [12,13]. The post-2011 predictions were more aligned to *in vitro* and mechanistic-based methods compared with pre-2011. All administration routes and modalities were included in the analysis. One hundred and thirteen (97%) of the candidate drugs in the dataset were small molecules; 97 (84%) were delivered via the oral route; 7% via the IV route; and 9% via inhalation (Table 1). Inhaled drugs were assessed on systemic exposure only, and not local delivery to lung tissue.

We believe that it is important to consider both the quantitative and the qualitative aspects of a human PK prediction, as there are some aspects of the utility of a human PK prediction in a drug discovery setting that are qualitative and not captured by the purely objective criteria on parameter accuracy (Figure 1E). For example, some PK processes are difficult to predict quantitatively into human (e.g., dose relationship with gut efflux). Hence, the evaluation of the PK prediction performed here also incorporated any qualitative predictions, drug–drug interaction potential or uncertainties that were highlighted at the time of candidate nomination and recognized as being a potential risk to the progress of the compound in development. Consideration was also given to any specific PK aims of the project identified at nomination, for example, if it was desired to have a short or long $t_{1/2}$, or if a candidate best-in-class compound was designed to improve upon particular PK properties of a first-in-class compound, then more weight was added to the evaluation of prediction accuracy of the critical PK parameters. Finally, qualitative feedback from the drug development project was taken to determine whether there were any significant challenges and/or delays related to PK during the **ascending dose trials**, which could be aligned to a misprediction (or lack of understanding of impact) of a specific PK property.

The combined quantitative plus qualitative assessment (Table 1) led to a classification of the PK predictions for each candidate drug in one of the following categories:

- **Successful:** most parameters were well predicted (within twofold of observed), and the overall PK of the candidate drug predicted with good accuracy, requiring no changes to the clinical study plan.
- **Marginal:** one or more parameters poorly predicted (>5 -fold from observed) and therefore the overall PK profile may not be as expected. A high level of uncertainty in a predicted parameter, and the kinetic consequences may have been highlighted at nomination. Despite the misprediction of one or more parameters, the project did not have to significantly change clinical plans and was not adversely delayed.
- **Unsuccessful:** one or more parameters poorly predicted (>5 -fold from observed) resulting in a PK profile that was significantly different from expected, meaning clinical studies may have been changed, project delayed, or even stopped.

Incorporating qualitative aspects into the evaluation allowed more focus on mispredictions with greater adverse impact on drug development, but also introduced subjectivity into the process. In order to minimize the effect of any subjective bias, the predictions and risk assessment at

Key Figure

Overview of Human PK Prediction and Evaluation Methods

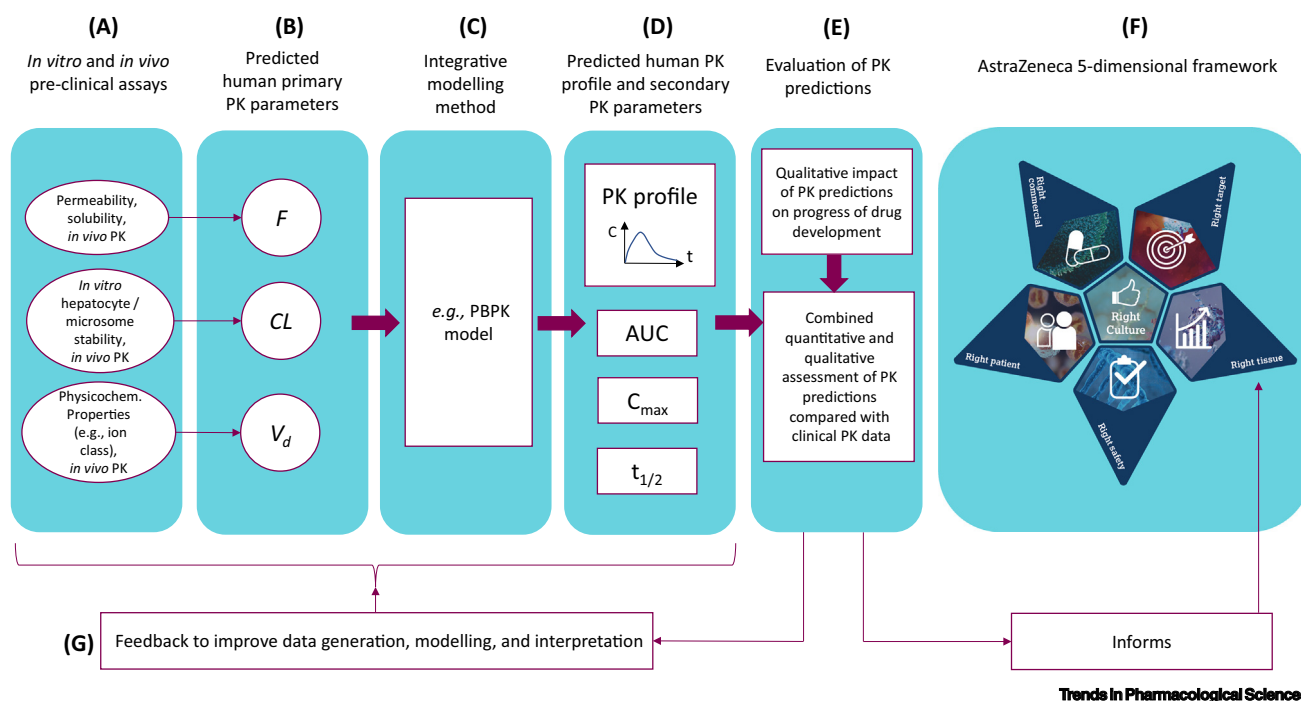


Figure 1. *In vitro* and *in vivo* ADME assays (A) are used to predict primary PK parameters F , CL , V_d (B), which can be used to build a PBPK model (C) for a candidate drug. The model can be used to generate the predicted human PK profile and secondary PK parameters AUC, C_{max} , and $t_{1/2}$ (D). The PK predictions are evaluated through a quantitative comparison of the secondary PK parameters with clinical PK data for the candidate drug, together with an assessment of the qualitative impact of PK predictions on the development of the candidate drug. PK predictions and their evaluation inform the Right Tissue/Exposure aspect of the AstraZeneca five-dimensional framework, alongside Right Target, Right Patients, Right Safety, and Right Commercial Potential (F), and provide feedback to improve data generation, interpretation and modelling involved in human PK prediction (G). Abbreviations: ADME, absorption, distribution, metabolism, and excretion; AUC, area under the curve; CL , clearance; C_{max} , maximum concentration; F , bioavailability; PBPK, physiologically based pharmacokinetic; PK, pharmacokinetic; $t_{1/2}$, half-life; V_d , volume of distribution.

the time of compound nomination were peer-reviewed by a panel of DMPK experts from all therapy areas within AstraZeneca, to ensure that consistent benchmarking was applied and that a consensus view was achieved for the more subjective elements of the evaluation. For example,

Box 1. Inhalation PBPK Models Developed by AstraZeneca

We have developed two PBPK models to predict human blood and lung PK profiles and explore systemic and pulmonary PK–PD relationships. The first model is a semiphysiological model that describes the PK behavior of soluble drugs through the rate of unbound drug transfer between five compartments [65] (Figure 1A). A main lung compartment is the dosing compartment for inhaled drug administration and a second lung compartment describes a subcompartment within the lung. The unbound CL from this compartment is slow and can be thought of as defining the process of lysosomal trapping (of drugs with ionizable basic centers) as a result of the pH difference between cytosol and lysosomes. A central compartment represents the systemic blood and two further compartments are for drug distribution in all other tissues (these are identical in structure to the lung compartments). The model has been shown to allow translation of lung PK between animal species and to effectively predict drug concentrations in human lung that correlate with efficacy for different classes of bronchodilators [65]. The model has advantages in simplicity of data inputs and model description, but it does not inform of more complex processes involved with inhaled drug disposition or of unbound drug availability in different regional locations within the lung. This is useful for determining whether an inhaled drug will have effective duration at the site of pharmacological action within the different cells of the lung. For these purposes a more sophisticated PBPK modeling approach is used where regional drug deposition, dissolution and mucociliary clearance are included in a true systems pharmacology approach [63,64] (Figure 1B). In this second model the lung is divided into 24 airway generations (airway branching) and these in turn divided into epithelial lining fluid, epithelial, and subepithelial regions. The tracheobronchial region is perfused by the bronchial blood flow, and the alveolar region by the entire cardiac output from the pulmonary blood supply and perfusion-rate-limited distribution is assumed to apply for all tissues. Inhaled drug particle size distribution is physically determined and particles can be deposited in the extrathoracic, tracheobronchial, and alveolar regions. The model is used to describe unbound drug concentration in all regions of the lung and this is contextualized against cell potency data to predict efficacy, but also captures PK and effects in the rest of the body.

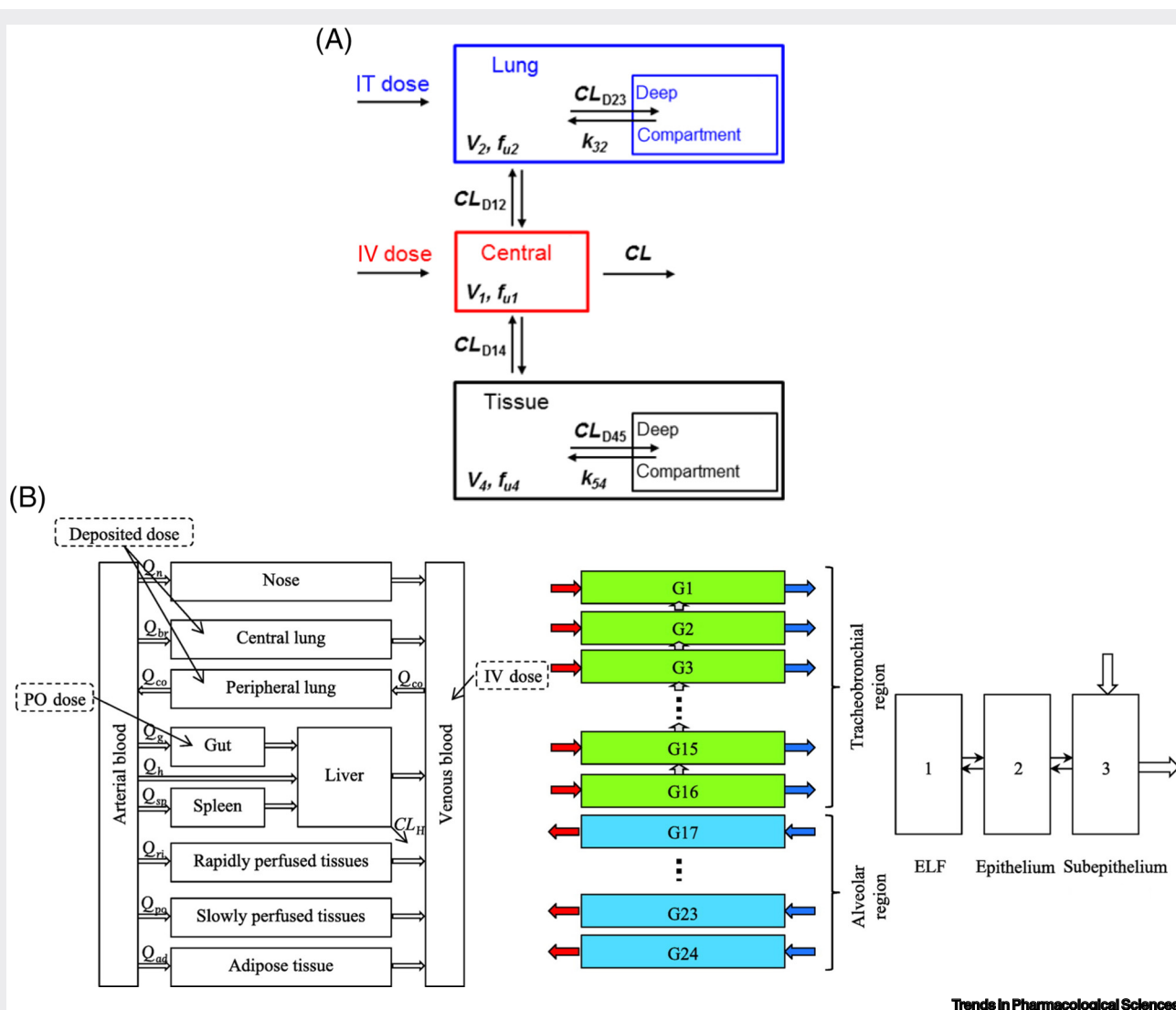


Figure 1. Physiologically Based Pharmacokinetic (PK) Modelling Approaches Used for Inhaled Drug Projects. (A) A semiphysiological compartmental model capturing key features of lung and systemic PK after administration of soluble drugs. (B) A fully physiologically based PK model to predict local unbound drug concentration and response in all relevant effect compartments on the lung. Abbreviations: CL, clearance; ELF, epithelial lining fluid; IT, intratracheal; IV, intravenous; PO, per os (oral).

if the compound under consideration was in development by the Cardiovascular, Renal, and Metabolic R&D organization, then it could be reviewed by DMPK experts from the Oncology R&D and Respiratory, Inflammation, and Autoimmune R&D organizations, who had not been involved in optimization or nomination of the compound.

Improved Prediction Methods Have Increased Prediction Accuracy

Since 2011 when the AstraZeneca five-dimensional framework was launched [12,13] (Figure 1F), 83% of compound predictions were assessed as successful based on the overall assessment of quantitative prediction accuracy plus qualitative project impact as opposed to 57% in 2000–2010 (Figure 2). Furthermore, 10% were marginal and 7% unsuccessful in 2011–2018, compared with 25% marginal, and 17% unsuccessful for predictions in 2000–2010 (Figure 2). This shows an

Table 1. PK Prediction Assessment Results for All AstraZeneca Compounds Evaluated

ID	Year FTIH	Target class	Therapy area	Route of administration	Compound type	Ion class	Overall combined quantitative plus qualitative evaluation result	Predicted/observed AUC ^a	Predicted/observed C _{max}	Predicted/observed t _{1/2}
1	1999	Nuclear receptor	Cardiovascular	PO	Small molecule	Acid	Successful			
2	2000	GPCR ^b	Cardiovascular	PO	Small molecule	Neutral	Marginal	2.50		2.0
3	2001	GPCR	Gastrointestinal	PO	Small molecule	Zwitterion	Unsuccessful	0.20		1.04
4	2002	Kinase	Oncology	PO	Small molecule	Base	Marginal			
5	2002	Ion channel	Cardiovascular	PO	Small molecule	Base	Successful	1.63		
6	2002	Ion channel	Gastrointestinal	PO	Small molecule	Neutral	Successful	1.49		0.40
7	2002	Other	Infection	IV	Small molecule	Acid	Successful	1.10		
8	2002	GPCR	Infection	PO	Small molecule	Acid	Successful			
9	2002	Ion channel	Respiratory	PO	Small molecule	Base	Successful	1.12		1.78
10	2002	Other enzyme	Respiratory	PO	Small molecule	Neutral	Successful			
11	2003	Kinase	Oncology	PO	Small molecule	Base	Successful	1.16	1.00	2.05
12	2003	Kinase	Oncology	PO	Small molecule	Base	Marginal	0.82		
13	2003	Kinase	Oncology	PO	Small molecule	Neutral	Unsuccessful	4.13	0.33	19.13
14	2003	Protease	Cardiovascular	PO	Small molecule	Acid	Unsuccessful	5.67	3.08	0.60
15	2003	Protease	Cardiovascular	PO	Small molecule	Neutral	Successful	1.07		
16	2003	Ion channel	Neurology	PO	Small molecule	Neutral	Unsuccessful			
17	2003	GPCR	Respiratory	PO	Small molecule	Acid	Unsuccessful	14.68	1.47	5.45
18	2003	GPCR	Respiratory	PO	Small molecule	Base	Marginal	1.02	0.39	0.26
19	2003	GPCR	Respiratory	PO	Small molecule	Neutral	Unsuccessful	1.35		
20	2004	Kinase	Oncology	PO	Small molecule	Neutral	Successful	3.49	1.08	1.58
21	2004	Ion channel	Cardiovascular	PO	Small molecule	Acid	Successful			
22	2004	Nuclear receptor	Cardiovascular	PO	Small molecule	Acid	Successful	0.52		
23	2004	Nuclear receptor	Cardiovascular	PO	Small molecule	Acid	Successful	0.59		
24	2004	GPCR	Gastrointestinal	PO	Small molecule	Zwitterion	Successful	0.96		
25	2004	Protease	Respiratory	PO	Small molecule	Neutral	Successful	1.53	1.58	1.79
26	2004	Protease	Respiratory	PO	Small molecule	Neutral	Successful	0.69	0.43	2.0
27	2004	GPCR	Respiratory	PO	Small molecule	Zwitterion	Marginal	1.0		2.0
28	2005	Kinase	Oncology	PO	Small molecule	Neutral	Successful	0.64	0.95	0.63
29	2005	Nuclear receptor	Cardiovascular	PO	Small molecule	Acid	Marginal	0.09		
30	2005	Kinase	Neurology	PO	Small molecule	Base	Successful	0.63	0.87	0.84
31	2005	GPCR	Neurology	PO	Small molecule	Neutral	Successful	0.56	0.75	0.33
32	2005	GPCR	Respiratory	PO	Small molecule	Acid	Unsuccessful	11.50		2.40
33	2006	Kinase	Oncology	PO	Small molecule	Base	Marginal	2.50	1.06	1.25

34	2006	Other	Oncology	IV	Small molecule	Base	Successful	0.40	0.79	0.61
35	2006	Kinase	Oncology	IV	Small molecule	Zwitterion	Marginal	0.94	0.33	4.80
36	2006	GPCR	Cardiovascular	PO	Small molecule	Neutral	Unsuccessful	8.93	0.38	1.38
37	2006	GPCR	Cardiovascular	PO	Small molecule	Neutral	Successful	0.60	0.26	0.61
38	2006	Other enzyme	Infection	PO	Small molecule	Acid	Unsuccessful	5.81	1.06	0.43
39	2006	GPCR	Neurology	PO	Small molecule	Base	Successful	1.11		0.54
40	2006	GPCR	Neurology	PO	Small molecule	Base	Successful	1.29		0.65
41	2006	Other enzyme	Neurology	PO	Small molecule	Neutral	Marginal		1.17	2.19
42	2006	Kinase	Respiratory	Inhaled	Small molecule	Base	Successful	0.80	0.93	0.88
43	2006	GPCR	Respiratory	PO	Small molecule	Base	Successful	1.97	1.37	1.04
44	2006	GPCR	Respiratory	Inhaled	Small molecule	Base	Marginal	4.0	2.0	0.45
45	2007	Kinase	Oncology	IV	Small molecule	Base	Successful	0.99	0.86	0.87
46	2007	Kinase	Oncology	PO	Small molecule	Neutral	Marginal	8.05	3.31	0.88
47	2007	Kinase	Cardiovascular	PO	Small molecule	Neutral	Successful	1.53	1.12	0.63
48	2007	GPCR	Cardiovascular/renal/ gastrointestinal	PO	Small molecule	Acid	Unsuccessful	168.0	41.61	1.63
49	2007	Ion channel	Cardiovascular/renal/ gastrointestinal	PO	Small molecule	Base	Successful	0.86		
50	2007	GPCR	Neurology	PO	Small molecule	Acid	Successful	2.0	0.93	2.30
51	2007	Other enzyme	Neurology	PO	Small molecule	Neutral	Successful	1.07	1.16	
52	2007	GPCR	Neurology	PO	Small molecule	Neutral	Successful	0.54	1.47	0.29
53	2007	Ion channel	Neurology	PO	Small molecule	Neutral	Marginal	0.17	0.25	0.39
54	2007	GPCR	Neurology	PO	Small molecule	Neutral	Unsuccessful	0.07		0.03
55	2007	Ion channel	Neurology	PO	Small molecule	Neutral	Successful	1.23	0.33	
56	2007	Ion channel	Neurology	PO	Small molecule	Neutral	Successful	0.44	0.21	0.47
57	2007	GPCR	Respiratory	PO	Small molecule	Base	Unsuccessful	1.66	0.54	2.94
58	2007	GPCR	Respiratory	Inhaled	Small molecule	Base	Successful	0.55	1.76	0.62
59	2007	Protease	Respiratory	PO	Small molecule	Neutral	Successful	1.12	1.03	1.05
60	2007	Protease	Respiratory	Inhaled	Small molecule	Neutral	Successful	1.24		1.29
61	2008	Kinase	Oncology	PO	Small molecule	Neutral	Successful	0.63	0.29	1.18
62	2008	Other enzyme	Gastrointestinal	PO	Small molecule	Acid	Marginal	3.53	1.54	0.45
63	2008	Kinase	Gastrointestinal	PO	Small molecule	Neutral	Successful	1.73	0.57	1.83
64	2008	Ion channel	Neurology	PO	Small molecule	Base	Successful			
65	2008	GPCR	Neurology	PO	Small molecule	Base	Marginal	5.37	3.60	1.30
66	2008	GPCR	Neurology	PO	Small molecule	Base	Marginal	0.39	0.81	0.32
67	2008	GPCR	Neurology	PO	Small molecule	Neutral	Marginal	0.27	0.31	0.36

(continued on next page)

Table 1. (continued)

ID	Year FTIH	Target class	Therapy area	Route of administration	Compound type	Ion class	Overall combined quantitative plus qualitative evaluation result	Predicted/observed AUC ^a	Predicted/observed C _{max}	Predicted/observed t _{1/2}
68	2008	GPCR	Respiratory	PO	Small molecule	Acid	Successful	0.72		0.73
69	2008	GPCR	Respiratory	PO	Small molecule	Acid	Unsuccessful	5.17	4.86	0.97
70	2008	GPCR	Respiratory	PO	Small molecule	Base	Successful	2.24	0.90	1.60
71	2009	Kinase	Oncology	PO	Small molecule	Base	Marginal	0.48	1.46	0.31
72	2009	Other enzyme	Cardiovascular	PO	Small molecule	Acid	Successful	0.79	0.33	0.92
73	2009	Other enzyme	Cardiovascular/renal/gastrointestinal	PO	Small molecule	Acid	Successful	0.38	0.35	1.32
74	2009	Kinase	Gastrointestinal	PO	Small molecule	Neutral	Marginal	0.24	0.12	0.54
75	2009	GPCR	Neurology	PO	Small molecule	Base	Unsuccessful	3.69	1.09	0.34
76	2009	GPCR	Neurology	PO	Small molecule	Base	Successful	1.14	0.81	1.16
77	2009	GPCR	Respiratory	PO	Small molecule	Acid	Successful	2.0	0.70	1.88
78	2009	GPCR	Respiratory	PO	Small molecule	Acid	Unsuccessful	4.03	1.33	2.22
79	2009	GPCR	Respiratory	Inhaled	Small molecule	Cationic	Marginal	0.60	0.14	0.10
80	2009	GPCR	Respiratory	Inhaled	Small molecule	Cationic	Marginal	4.18	2.0	0.93
81	2009	Protease	Respiratory	PO	Small molecule	Neutral	Marginal	0.64	0.14	2.45
82	2009	Nuclear receptor	Respiratory	Inhaled	Small molecule	Neutral	Successful	1.03	0.67	0.69
83	2010	Nuclear receptor	Oncology	PO	Small molecule	Base	Successful	0.71	0.43	1.85
84	2010	Kinase	Oncology	PO	Small molecule	Base	Successful	2.50		0.63
85	2010	Kinase	Oncology	PO	Small molecule	Neutral	Successful	0.47	0.24	3.20
86	2010	Kinase	Oncology	PO	Small molecule	Neutral	Marginal	0.25	0.12	1.49
87	2010	Kinase	Cardiovascular	PO	Small molecule	Neutral	Successful	1.86	1.12	0.86
88	2012	Kinase	Oncology	PO	Small molecule	Base	Unsuccessful			
89	2012	Kinase	Respiratory	Inhaled	Small molecule	Base	Successful	1.40	0.88	0.51
90	2012	Nuclear receptor	Respiratory	Inhaled	Small molecule	Neutral	Successful	1.0	1.0	
91	2013	Kinase	Oncology	PO	Small molecule	Base	Marginal	0.15	0.58	0.24
92	2013	Kinase	Oncology	PO	Small molecule	Neutral	Successful	0.23	0.26	0.67
93	2013	Other	Oncology	PO	Small molecule	Neutral	Successful	0.95	1.11	0.76
94	2013	Kinase	Oncology	PO	Small molecule	Neutral	Successful	2.91	1.25	1.38
95	2013	GPCR	Respiratory	PO	Small molecule	Acid	Successful	1.14	0.73	1.67
96	2014	Kinase	Oncology	PO	Small molecule	Base	Successful	2.56		0.94
97	2014	Kinase	Oncology	PO	Small molecule	Neutral	Successful	0.41	0.34	1.37
98	2014	Nuclear receptor	Oncology	PO	Small molecule	Zwitterion	Unsuccessful	7.73	1.28	4.35
99	2014	GPCR	Cardiovascular/renal/	PO	Small molecule	Base	Successful	1.71	1.23	0.83

			gastrointestinal							
100	2014	Protease	Respiratory	PO	Small molecule	Base	Successful	0.92	0.59	0.80
101	2015	Kinase	Oncology	PO	Small molecule	Base	Marginal	7.07	3.25	1.05
102	2015	Kinase	Oncology	IV	Nanomedicine	Base	Successful	0.83	0.88	1.59
103	2016	GPCR	Oncology	PO	Small molecule	Neutral	Successful	1.07	1.33	0.24
104	2016	Nuclear receptor	Cardiovascular	PO	Small molecule	Neutral	Successful	1.69	1.51	0.25
105	2016	Other	Cardiovascular	PO	Small molecule	Neutral	Successful	2.99		0.56
106	2016	Ion channel	Respiratory	Inhaled	Small molecule	Base	Successful	0.86	0.58	0.53
107	2016	Nuclear receptor	Respiratory	PO	Small molecule	Neutral	Successful	0.81	0.41	1.19
108	2016	Nuclear receptor	Respiratory	PO	Small molecule	Neutral	Successful	0.40	0.64	1.00
109	2017	Other	Oncology	IV	Small molecule	Acid	Successful	1.64	0.67	0.46
110	2017	Other	Oncology	PO	Small molecule	Base	Successful	1.26	1.02	0.67
111	2017	Other enzyme	Oncology	IV	Oligonucleotide	N/A	Successful	0.77	1.00	1.06
112	2017	Other enzyme	Cardiovascular	PO	Small molecule	Base	Successful	0.90		0.28
113	2017	Other enzyme	Respiratory	PO	Small molecule	Acid	Successful	0.77	0.67	1.51
114	2018	Kinase	Oncology	PO	Small molecule	Base	Successful	1.20	2.66	1.44
115	2018	Kinase	Oncology	IV	Small molecule	Neutral	Marginal	0.30		0.40
116	2018	Other	Respiratory	Inhaled	Anticalin	N/A	Successful	0.80		0.40

^aMissing AUC, $C_{\max}$, and $t_{1/2}$ ratios due to unavailability of either predicted or observed values for these quantities.

^bAbbreviations: FTIH, first-time-in-human (dosing); GPCR, G-protein-coupled receptor; ID, identifier number; PO, *per os* (oral dosing).

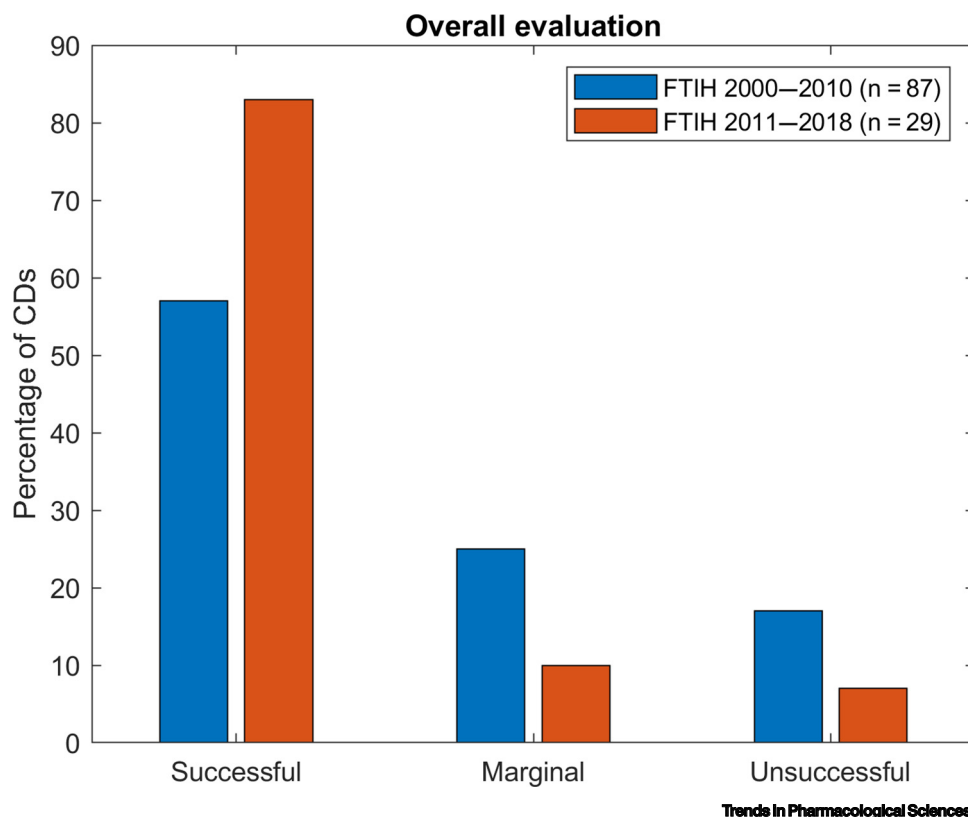
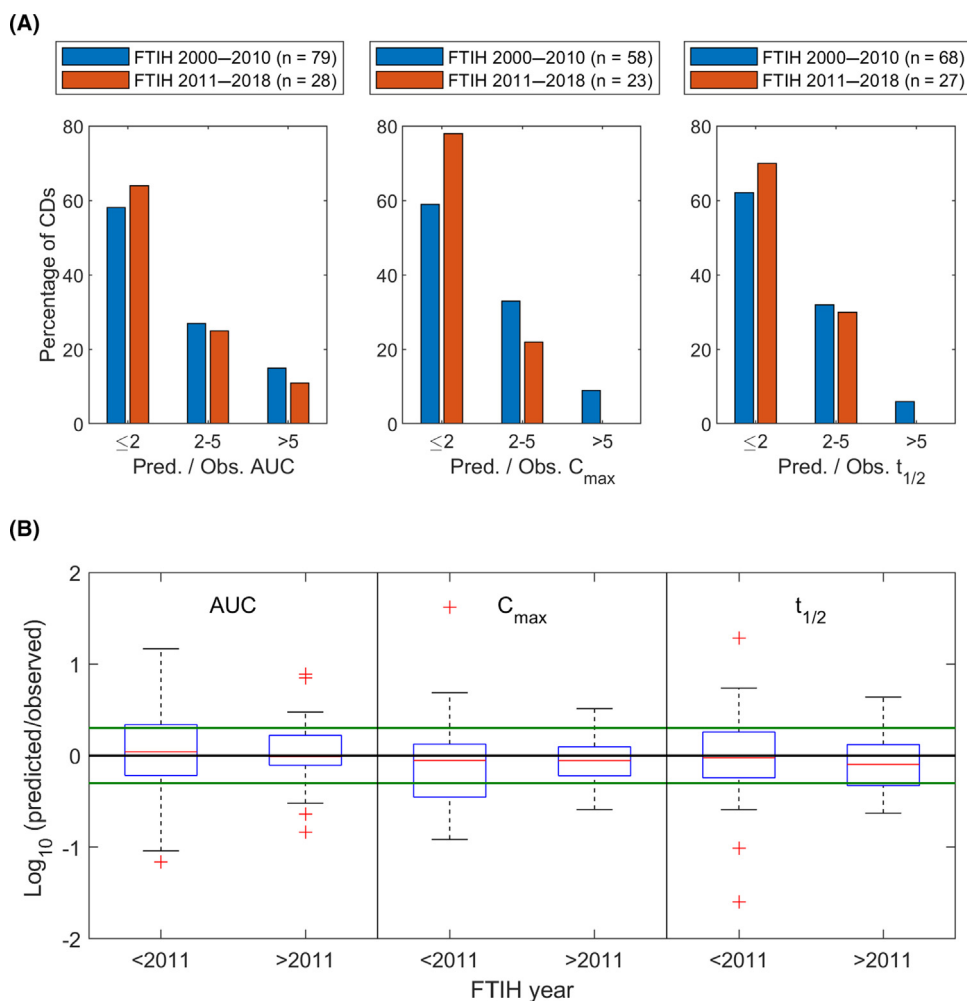


Figure 2. Overall Evaluation Incorporating Both Quantitative and Qualitative Aspects of Pharmacokinetic (PK) Predictions, for Candidate Drugs (CDs) That Were Dosed First-Time-in-Human (FTIH) in 2000–2010 and 2011–2018.

increase over time in the frequency of successful predictions, and a corresponding decrease in the proportion of marginal or unsuccessful predictions. It is important to note that the 7% of unsuccessful predictions since 2011 represent just two compounds, previously discussed in [13].

The quantitative prediction accuracy of PK parameters showed a trend to increase in the period since 2011 (Figure 3). Predictions of $C_{\max}$ were most frequently accurate with 78% (18/23) within twofold, followed by $t_{1/2}$ with 70% (19/27) within twofold, and AUC with 64% (18/28) within twofold. For all three parameters, the percentage accurately predicted within twofold was higher post-2011, and the percentage predicted outside fivefold was lower post-2011 (Table 2 and Figure 3A). Notably, none of the $C_{\max}$ or $t_{1/2}$ predictions in the period since 2011 were more than fivefold from the observed values. In total, 71% (55/78) of the predicted/observed parameter ratios were within twofold during 2011–2018. Furthermore, the median predicted/observed ratios for AUC, $C_{\max}$, and $t_{1/2}$ are close to 1 (Figure 3B), demonstrating that there was no overall trend to over- or underpredict these parameters. The box plots also show that the variation of the predicted/observed ratios of the PK parameters, depicted as interquartile range, was smaller for the compound predictions since 2011, meaning that a greater proportion of the predicted parameters were close to the observed values.

We compared the prediction accuracy from our AZ dataset (presented in Figure 3A) with other reviews of PK methodology; Zhang *et al.* [66], Van den Bergh *et al.* [67], and Jones *et al.* [68]



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Figure 3. Accuracy of PK Parameter Predictions for Candidate Drugs That Were Dosed First-Time-in-Human (FTIH) in 2000–2010 and 2011–2018. (A) Graphical representation showing accuracy of different quantitative PK predictions parameters (AUC, C_{max} , and $t_{1/2}$) for AstraZeneca CDs that were dosed FTIH in 2000–2010 (blue) and 2011–2018 (red). (B) Box plots of the ratio of predicted to observed PK parameter values for AstraZeneca CDs that were dosed FTIH in 2000–2010 and 2011–2018. Positive log ratios show overprediction (actual values lower than predicted) and negative log ratios underprediction (actual values higher than predicted). The black line shows a ratio equal to 1 (predicted value = observed value), and green lines show predicted values that are double or half the observed value. Red line shows median, box shows interquartile range, and crosses are outliers. Abbreviations: AUC, area under the curve; CD, candidate drug; C_{max} , maximum concentration; FTIH, first-time-in-human; PK, pharmacokinetic; $t_{1/2}$, half-life.

(Table 1). Our analysis shows prediction accuracy for all parameters that is at least as good as, if not superior to these previous analyses, with a substantially larger sample size.

Predictions Are More Accurate for Basic Compounds Than Neutral or Acidic Compounds

The benefit of having reviewed a large number of compound predictions is that we can examine the prediction accuracy for subsets of the data; for example, for different ion classes. The AstraZeneca dataset (Table 1) comprises 45 neutral compounds, 39 bases, 23 acids, five zwitterions, two cations, and two compounds for which an ion class cannot be assigned. We

Table 2. Comparison of Percentages of AstraZeneca CDs with Predictions within Twofold of Observed Parameter Values (AUC, $C_{\max}$, and $t_{1/2}$) with Other Reported Works

Evaluation	% CDs ^b with AUC predicted within twofold	% CDs ^b with $C_{\max}$ predicted within twofold	% CDs ^b with $t_{1/2}$ predicted within twofold
AZ 2000–2010	58% (46/79)	59% (34/58)	62% (42/68)
AZ 2011–2018	64% (18/28)	78% (18/23)	70% (19/27)
Van den Bergh <i>et al.</i> [67] ^a	26–51%	46–63%	43–60%
Jones <i>et al.</i> [68]	50% (9/18)	67% (12/18)	61% (11/18)
Zhang <i>et al.</i> [66]	63% (10/16)	88% (14/16)	69% (11/16)

^aResults given as ranges due to evaluation of a variety of methods ($n = 35$ CDs).^bAbbreviation: CD, candidate drug.

found that more parameters were predicted within twofold of observed values for basic compounds than neutrals or acids, especially for $C_{\max}$ (Figure 4A). For neutral compounds, it was challenging to predict $C_{\max}$ (with an overall bias for underprediction; Figure 4B), potentially due to complex dissolution processes, but AUC and $t_{1/2}$ were accurately predicted. For acidic compounds, it was challenging to predict $t_{1/2}$ or AUC, with nearly a third of AUC predictions >5-fold different to the observed value, and a bias for overprediction by about twofold (Figure 4B). Discrepancies for acids may be due to complex oral dissolution profiles; CL /transport mechanisms not captured by *in vitro* assays (e.g., UGT); or the sensitivity of $t_{1/2}$ accuracy to CL mispredictions due to their low V_d . In all, human PK predictions for bases were consistently found to be reliable, whereas for neutral compounds $C_{\max}$ was underestimated in many cases, and for acids both AUC and $t_{1/2}$ often were more frequently mispredicted.

Concluding Remarks and Future Perspectives

Retrospective evaluation of human PK prediction is essential for improving prediction methods and strategies for future candidate drugs (Figure 1E,G). The current state-of-the-art methods for small-molecule drugs were developed and refined by learning from unsuccessful predictions. However, it is important to avoid a black-box approach with these tools and methods. The philosophy of PK prediction at AstraZeneca values knowledge gaining and truth-seeking behaviors, while striving to minimize the risk of clinical failure of candidate drugs due to unacceptable exposure profiles. This contrasts with a more deterministic method-led approach implementing a naïve algorithmic translation of *in vitro* and *in vivo* data to humans, which, in the authors' opinion, can lead to poor decisions and outcomes, if a thorough mechanistic understanding of the compound's disposition properties is missing. Thus, it is necessary but not sufficient for PK scientists in AstraZeneca to provide project teams with data from the plethora of PK assays and models available to projects in most pharmaceutical companies. Moreover, their role, working in partnership with Medicinal Chemists and Drug Design teams, is to develop a quantitative mechanistic understanding of the disposition properties of key compounds via detailed experimentation and thorough investigations towards optimizing and predicting the key human parameters that support oral PK profiles (F_a , V_d , and CL), underpinned by a knowledge of the exposure–effect relationship. Applying this philosophy, supported with data from the best practice PK prediction methods, has allowed us to improve the accuracy of human PK predictions since 2011 to deliver the Right Tissue/Exposure considerations of AstraZeneca's five-dimensional framework for drug development.

While not unique to AstraZeneca, the relative success of achieving acceptable exposure profiles for the majority of candidate drugs has been underpinned by the following complementary

Outstanding Questions

How should the uncertainty in a PK prediction be calculated and expressed?

How can we improve the accuracy of prediction of interindividual variability, including PK in special populations?

How can we utilize advanced algorithms, such as deep learning techniques, to significantly improve the prediction of human PK profiles from chemical structure alone?

What are the best practices for predicting human PK from nonclinical data for the broad range of new modality therapies being developed?

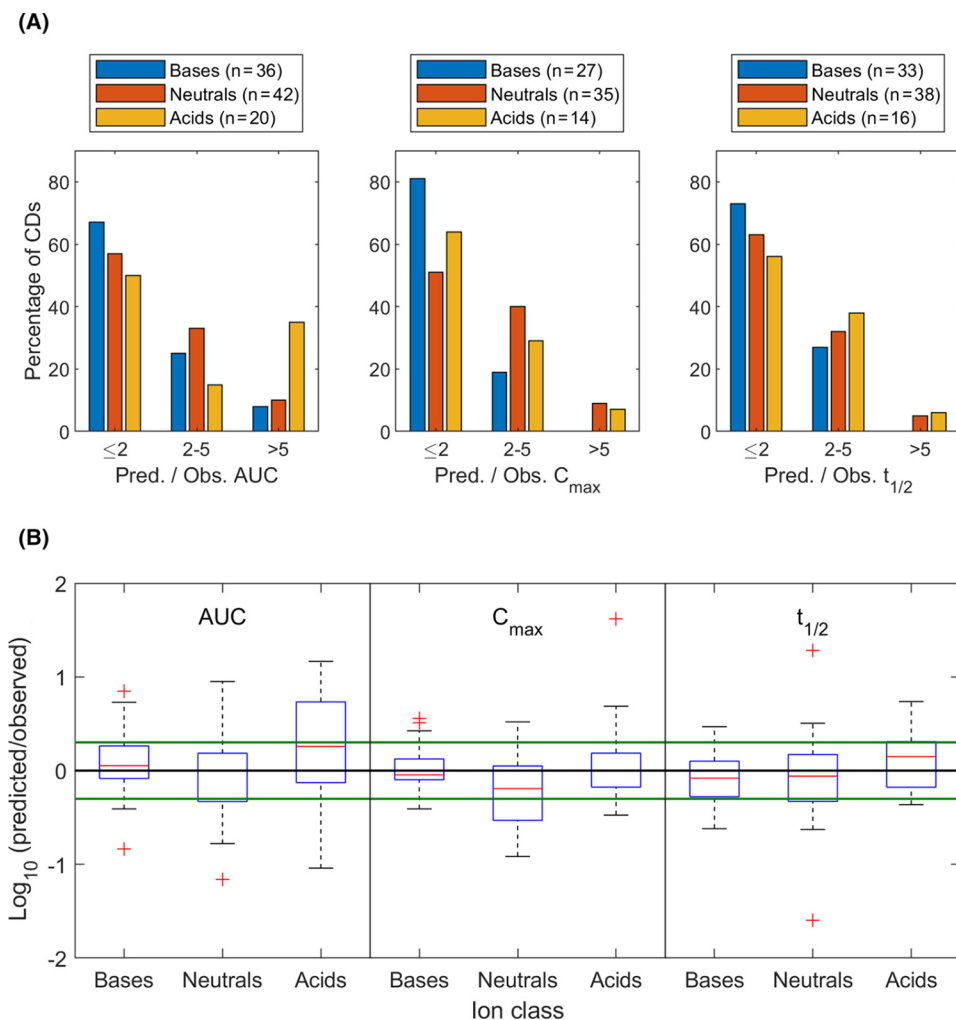


Figure 4. Accuracy of PK Parameter Predictions for Basic, Neutral, or Acidic Compounds. (A) Graphical representation showing accuracy of different quantitative PK predictions of parameters (AUC, C_{max} , and $t_{1/2}$) for basic (blue), neutral (red), or acidic (yellow) AstraZeneca CDs. (B) Box plots of the ratio of predicted to observed PK parameter values for basic, neutral, or acidic compounds. Positive log ratios show overprediction (actual values lower than predicted) and negative log ratios underprediction (actual values higher than predicted). The black line shows a ratio equal to 1 (predicted value = observed value), and green lines show predicted values that are double or half the observed value. Red line shows median, box shows interquartile range, and crosses are outliers. Abbreviations: AUC, area under the curve; CD, candidate drug; C_{max} , maximum concentration; PK, pharmacokinetic; Pred, predicted; $t_{1/2}$, half-life.

approaches: achieving an understanding of the interdependencies between physicochemical properties and key PK parameters; consideration of data, information, and knowledge from well-validated *in vitro* assays with human-derived material and *in vivo* animal models; and knowledge gained from tailored experimentation to better define key risks associated with a chemical series and specific compounds.

Given the complexities of ADME processes in humans, especially absorption and metabolism, it may not be possible to increase the pure quantitative accuracy of PK predictions significantly beyond what has been presented here; however, there are other areas in which prediction methods

can be refined and expanded (see [Outstanding Questions](#)). Some approaches to estimate the uncertainty of PK predictions have been proposed. Several authors have considered this issue and provided advice on good practice to ensure uncertainty is taken into account, however, there is no single approach that is agreed best practice. This is perhaps because the challenge is as much about how to communicate the prediction uncertainty in a manner that is useful for internal company decision makers and external regulators as it is about how best to perform the calculations. Continued dialogue and enhanced experience will help to establish a common language and best practice.

It is also important to consider the interindividual variability in human PK predictions, and PBPK tools such as Simcyp and GastroPlus [9,18] already allow simulation for different demographics. Recent developments in this area include the definition of population characteristics for special populations; for example, oncology patients, who have been shown to have differences in plasma protein binding and metabolism to healthy individuals [69,70]. Further data on demographic or disease variations need to be compiled to allow accurate simulation of PK in relevant populations as therapies become increasingly personalized.

The use of *in silico* PK predictions based solely on chemical structure could significantly and proactively aid smart chemical drug design before compounds are synthesized. Computational or regression methods to predict primary PK parameters from chemical structure have been investigated [71–74], and they are most accurate for distribution parameters such as V_d or f_{ub} , for which the dependence on physicochemical properties are already understood [75]. These approaches are derived from databases of PK parameters and chemical structure descriptors, and they are purely empirical, by contrast to the mechanistic methods described in this article, which may limit their predictive power. However, as the databases become larger and computational power makes machine learning methods more tractable, empirical predictions are likely to become more accurate, and could provide useful information, complementary to the mechanistically derived parameter values, to factor into human PK predictions. Predictions from complex regression methods should be interpreted with the same view of how it fits with the overall picture for the candidate drug, and of the method limitations, as with any other data source.

Since 2011, AstraZeneca's Research and Early Development Organization has increased the diversity of approaches to potential therapeutics in drug discovery, investigating several different new modalities [12,13], in addition to traditional small molecules. These are at various stages from early discovery identification to late stage clinical testing and present new challenges and ask different questions of PK scientists. Only some of the *in vitro* and *in vivo* ADME methods that are optimized for small molecules will be appropriate for new modalities. New assay methods need to be developed to fully understand the ADME characteristics of the larger, more complex molecules required to address important pharmacological targets where small molecule approaches have not been feasible. However, a knowledge-driven approach, understanding the mechanisms being represented by different assays, as well as the underlying assumptions and limitations, will remain essential to providing accurate predictions for new chemical modalities, as for small molecule candidate drugs.

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Disclaimer Statement

The authors are employees of AstraZeneca.

Resources

ⁱwww.phrma.orgⁱⁱwww.certara.com/pbpbk-modeling-and-simulation/?ap=PBPKⁱⁱⁱwww.simulations-plus.com/software/gastroplus/

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